

Dominic (Nick) Puglise

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OBJECTIVE

Junior Mechanical Engineering student at Milwaukee School of Engineering (MSOE). Experienced with interdisciplinary projects in both the mechanical and electrical engineering sector. Fast learner, motivated, and looking for an internship in mechanical engineering for fall/summer 2025.

EDUCATION

B.S. Mechanical Engineering | Milwaukee School of Engineering | Anticipated May 2026 | GPA 3.5
Highschool | Wheaton Academy | 2016-2020 | GPA 3.6

LEADERSHIP | CO-CURRICULAR INVOLVEMENT

Lead Design Engineer | FIRST Robotics | August 2016 – May 2020
Competitor | MSOE Esports (Super Smash Bros Ultimate) | September 2022 – Current
Member | MSOE Raider 3D | September 2023 – Current

SKILLS | PROGRAMS

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| ▪ Autodesk Fusion360 | ▪ MATLAB/Simulink | ▪ Arduino C++, Python, Javascript, R |
| ▪ SolidWorks | ▪ Adobe Product Suite | ▪ Formal and Technical Writing |
| ▪ Cura, CHITUBOX, Universal Gcode Sender | ▪ Microsoft Office Suite | ▪ Presentation Skills |
| | ▪ MIG/TIG Welding | ▪ EasyEDA |

EXAMPLE PROJECT EXPERIENCE

Electronics Projects | Personal Interest | 2015 - Current

Project Objective: This topic encompasses several personal electronic projects I have pursued in my spare time. A significant portion of my electrical projects are centered around functionally and aesthetically restoring vintage analog devices such old oscilloscopes, function generators, voltage/current/resistance meters, as well as consumer electronics such as CRT televisions, radios, and vintage computers.

- Requires electrical knowledge of the operating principles in both analog and digital electronics present in each device
- Involves extensive usage of 3D modeling for fabricating replacement parts from missing/damaged components

Pressure Gauge Project | Milwaukee School of Engineering | Milwaukee, WI | December 2023 – May 2024

Project Objective: The project aims to develop a usable pressure gauge for measuring compression in an internal combustion engine cylinder by utilizing strain measurement from elastic deformation of a metallic surface, processed through a Raspberry Pi.

- Pressure measurement is calculated from the circumferential elongation measured by a strain rosette gauge
- Requires use of machine shop equipment (Lathe, Mill, Bandsaw, Grinder, Drill, etc.)
- Electronics and calculations are handled by a raspberry pi, data is retrieved through serial communication and a web app hosted by the pi

Custom variable AC and DC power supply | Personal Interest | September 2023

Project Objective: The goal of this project was to develop an isolated variable AC and DC power supply system, with the Arduino controlling most functions. The outputs of this power supply range from 0-120 VAC and 1-160 V DC with a max current supply of 3.5 Amperes.

- The key electrical components include an isolation transformer, NPN transistor-driven relays, a resistive soft start transformer circuit, a variable autotransformer, a linear DC power supply circuit, voltage and current sensors, an I2C 16x2 LCD display, and an Arduino UNO that has been programmed in C++
 - Faceplate and associated CAM toolpath was modeled in Fusion360 and cut on a CNC router
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